Manav Rachna International Institute of Research and Studies

School of Computer Applications

PRACTICAL FILE

of

DATA STRUCTURE & ALGORITHM

Master of Computer Applications (MCA) – Semester 1

Academic Session 2026–2027

SUBMITTED BY	
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Course	: MCA, Semester 1
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Practical 1:

Basic Questions:

Question 1:
Program to calculate electricity bill with the following information:
- Units: 100 units, 101 to 300 units, above 300 units
- Respective prices: ₹3 per unit, ₹4 per unit, and ₹5 per unit

Question 2: Write a program to check whether a given number is even or odd.
Question 3: Write a program to find the largest and smallest of three numbers.
Question 4: Grade the following students with the following remarks:
- Students having marks 90 to 100 will be remarked as excellent.
- 70 to 90 to be remarked as very good.
- 60 to 70 to be remarked as good.
- 50 to 60 to be remarked as fair.
- Below 50 to be remarked as poor.

Question 5: Write a program for printing out tables from 1 to n,
where n is taken from the user as an input.

Code:

```
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 3
elif units <= 300:
    bill = 100 * 3 + (units - 100) * 4
else:
    bill = 100 * 3 + 200 * 4 + (units - 300) * 5

print(f"Total bill: ₹{bill}")
```

Screenshot:

```
# Question 1:
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 3
elif units <= 300:
    bill = 100 * 3 + (units - 100) * 4
else:
    bill = 100 * 3 + 200 * 4 + (units - 300) * 5

print(f"Total bill: ₹{bill}")
```

Code Output:

```
Enter units consumed: 300
Total bill: ₹1100
```

Question 2:

Code:

```
n = int(input("Enter a number: "))
print(f"{n} is", "even" if n % 2 == 0 else "odd")
```

Code Screenshot:

```
# Question 2:
n = int(input("Enter a number: "))
print(f"{n} is", "even" if n % 2 == 0 else "odd")
```

Output Screenshot:

```
Enter a number: 34567
34567 is odd
```

Question 3:

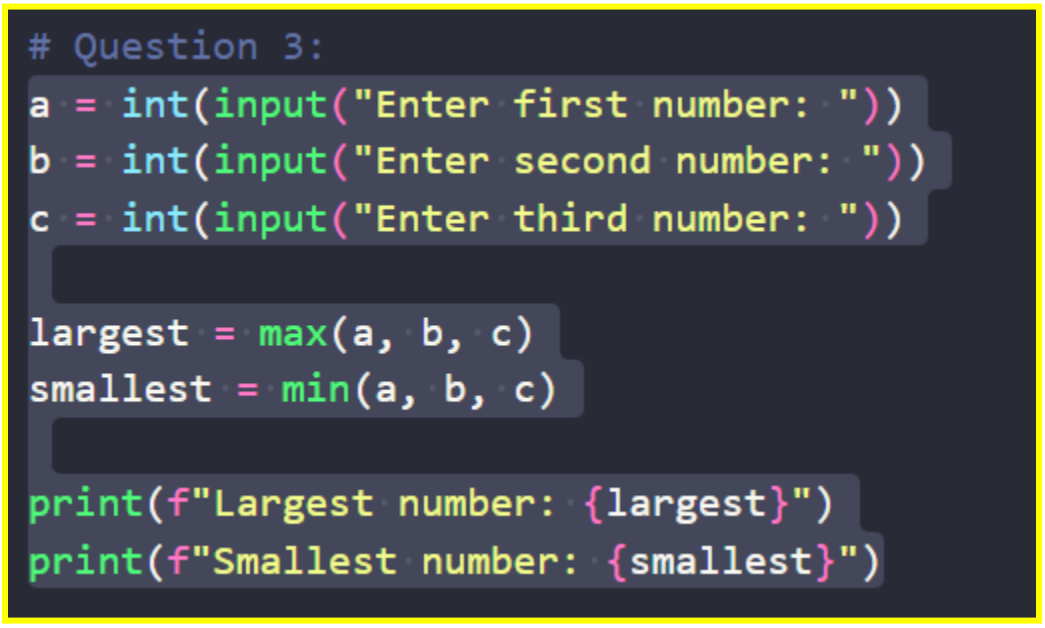
Code:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

largest = max(a, b, c)
smallest = min(a, b, c)

print(f"Largest number: {largest}")
print(f"Smallest number: {smallest}")
```

Code Screenshot:

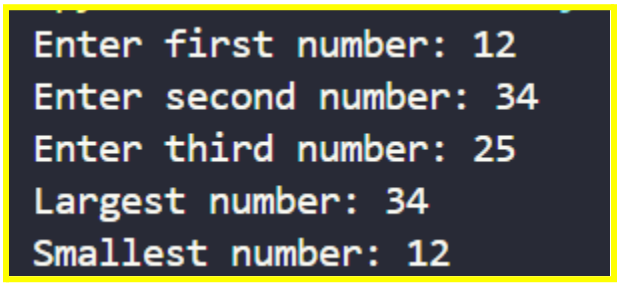


```
# Question 3:
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

largest = max(a, b, c)
smallest = min(a, b, c)

print(f"Largest number: {largest}")
print(f"Smallest number: {smallest}")
```

Code Output:



```
Enter first number: 12
Enter second number: 34
Enter third number: 25
Largest number: 34
Smallest number: 12
```

Question 4:

```
marks = float(input("Enter marks: "))

if 90 <= marks <= 100:
    print("Excellent")
elif 70 <= marks < 90:
    print("Very Good")
elif 60 <= marks < 70:
    print("Good")
elif 50 <= marks < 60:
    print("Fair")
else:
    print("Poor")
```

Code Screenshot:

```
# Question 4:
marks = float(input("Enter marks: "))

if 90 <= marks <= 100:
    print("Excellent")
elif 70 <= marks < 90:
    print("Very Good")
elif 60 <= marks < 70:
    print("Good")
elif 50 <= marks < 60:
    print("Fair")
else:
    print("Poor")
```

Code Output:

```
Enter marks: 87
Very Good
```

Question 5:

Code:

```
n = int(input("Enter the value of n: "))

for i in range(1, n + 1):
    print(f"Table of {i}:")
    for j in range(1, 11):
        print(f"{i} x {j} = {i * j}")
    print()
```

Code Screenshot:

```python
# Question 5:
n = int(input("Enter the value of n: "))

for i in range(1, n + 1):
    print(f"Table of {i}:")
    for j in range(1, 11):
        print(f"{i} x {j} = {i * j}")
    print()
```

Code Output:

```
Enter the value of n: 4
Table of 1:
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
1 x 5 = 5
1 x 6 = 6
1 x 7 = 7
1 x 8 = 8
1 x 9 = 9
1 x 10 = 10

Table of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
```

```
Table of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Table of 4:
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
```

Array Question:

Question 1. Insert a new element in the array.
Question 2: Delete an element from an array.
Question 3. Find the product of two matrices.
Question 4. Sum of diagonal and non-diagonal elements in a matrix.
Question 5. Enter marks of five students for three subjects each and find the percentage of students.

Question 1:

Code:

```
arr = list(map(int, input("Enter array elements: ").split()))
pos = int(input(f"Enter position (0 to {len(arr)}): "))
val = int(input("Enter element to insert: "))

if 0 <= pos <= len(arr):
    arr.insert(pos, val)
    print("Updated array:", arr)
else:
    print("Invalid position")
```

Code Screenshot:

```
# Question 1:
arr = list(map(int, input("Enter array elements: ").split()))
pos = int(input(f"Enter position (0 to {len(arr)}): "))
val = int(input("Enter element to insert: "))

if 0 <= pos <= len(arr):
    arr.insert(pos, val)
    print("Updated array:", arr)
else:
    print("Invalid position")
```

Code Output:

```
Enter array elements: 1
Enter position (0 to 1): 1
Enter element to insert: 3
Updated array: [1, 3]
```

Question 2:

Code:

```
arr = list(map(int, input("Enter array elements: ").split()))
pos = int(input(f"Enter position to delete (0 to {len(arr) - 1}): "))

if 0 <= pos < len(arr):
    removed = arr.pop(pos)
    print(f"Deleted {removed}. Updated array:", arr)
else:
    print("Invalid position")
```

Code Screenshot:

```
# Question 2:
arr = list(map(int, input("Enter array elements: ").split()))
pos = int(input(f"Enter position to delete (0 to {len(arr) - 1}): "))

if 0 <= pos < len(arr):
    removed = arr.pop(pos)
    print(f"Deleted {removed}. Updated array:", arr)
else:
    print("Invalid position")
```

Code Output:

```
Enter array elements: 5
Enter position to delete (0 to 0): 0
Deleted 5. Updated array: []
```

Question 3:

Code:

```
r1, c1 = map(int, input("Rows and columns of matrix A: ").split())
r2, c2 = map(int, input("Rows and columns of matrix B: ").split())

if c1 != r2:
```

```
    print("Multiplication not possible (columns of A must equal rows of B)")
else:
    print("Enter matrix A:")
    A = [list(map(int, input().split())) for _ in range(r1)]
    print("Enter matrix B:")
    B = [list(map(int, input().split())) for _ in range(r2)]

    C = [[0] * c2 for _ in range(r1)]
    for i in range(r1):
        for j in range(c2):
            for k in range(c1):
                C[i][j] += A[i][k] * B[k][j]

    print("Product:")
    for row in C:
        print(*row)
```

Code Screenshot:

```python
# Question 3:
r1, c1 = map(int, input("Rows and columns of matrix A: ").split())
r2, c2 = map(int, input("Rows and columns of matrix B: ").split())

if c1 != r2:
    print("Multiplication not possible (columns of A must equal rows of B)")
else:
    print("Enter matrix A:")
    A = [list(map(int, input().split())) for _ in range(r1)]
    print("Enter matrix B:")
    B = [list(map(int, input().split())) for _ in range(r2)]

    C = [[0] * c2 for _ in range(r1)]
    for i in range(r1):
        for j in range(c2):
            for k in range(c1):
                C[i][j] += A[i][k] * B[k][j]

    print("Product:")
    for row in C:
        print(*row)
```

Code Output:

```
Rows and columns of matrix A: 2 3
Rows and columns of matrix B: 2 5
Multiplication not possible (columns of A must equal rows of B)
```

Question 4:

Code:

```
n = int(input("Enter order of square matrix: "))
print("Enter matrix:")
M = [list(map(int, input().split())) for _ in range(n)]

diag = non_diag = 0
for i in range(n):
    for j in range(n):
        if i == j:
            diag += M[i][j]
        else:
            non_diag += M[i][j]

print("Sum of diagonal elements:", diag)
print("Sum of non-diagonal elements:", non_diag)
```

Code Screenshot:

```
# Question 4:
n = int(input("Enter order of square matrix: "))
print("Enter matrix:")
M = [list(map(int, input().split())) for _ in range(n)]

diag = non_diag = 0
for i in range(n):
    for j in range(n):
        if i == j:
            diag += M[i][j]
        else:
            non_diag += M[i][j]

print("Sum of diagonal elements:", diag)
print("Sum of non-diagonal elements:", non_diag)
```

Code Output:

```
Enter order of square matrix: 3
Enter matrix:
1 2 3
4 5 6
7 8 9
Sum of diagonal elements: 15
Sum of non-diagonal elements: 30
```

Question 5:

Code:

```
for s in range(1, 6):
    print(f"\nStudent {s}")
    total = 0
    for sub in range(1, 4):
        total += float(input(f"Marks in subject {sub}: "))
    percentage = total / 300 * 100
    print(f"Percentage: {percentage:.2f}%")
```

Code Screenshot:

```
# Question 5:
for s in range(1, 6):
    print(f"\nStudent {s}")
    total = 0
    for sub in range(1, 4):
        total += float(input(f"Marks in subject {sub}: "))
    percentage = total / 300 * 100
    print(f"Percentage: {percentage:.2f}%")
```

Code Output:

```
Student 1
Marks in subject 1: 34
Marks in subject 2: 56
Marks in subject 3: 78
Percentage: 56.00%

Student 2
Marks in subject 1: 97
Marks in subject 2: 45
Marks in subject 3: 90
Percentage: 77.33%

Student 3
Marks in subject 1: 67
Marks in subject 2: 87
Marks in subject 3: 89
Percentage: 81.00%
```

```
Student 4
Marks in subject 1: 45
Marks in subject 2: 37
Marks in subject 3: 87
Percentage: 56.33%

Student 5
Marks in subject 1: 98
Marks in subject 2: 97
Marks in subject 3: 99
Percentage: 98.00%
```

Stack Question:

Question 1: Write a A dynamic program for stack operations where the user can dynamically choose which operations to select.
the following are the options that need to be put forward:
1. Create a stack
2. Push the element into a stack
3. Pop an element
4. Display an element
5. Overflow condition
6. Underflow condition
7. Exit

Code:

```
stack = []
size = 0
created = False

def is_full():
    return len(stack) == size

def is_empty():
    return len(stack) == 0

while True:
    print("\n--- Stack Menu ---")
    print("1. Create a stack")
    print("2. Push an element")
    print("3. Pop an element")
    print("4. Display the stack")
    print("5. Overflow condition")
    print("6. Underflow condition")
    print("7. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        size = int(input("Enter size of the stack: "))
        if size <= 0:
            print("Size must be greater than 0")
        else:
            stack = []
```

```
            created = True
            print(f"Stack of size {size} created")

    elif choice in ("2", "3", "4", "5", "6") and not created:
        print("Create a stack first (option 1)")

    elif choice == "2":
        if is_full():
            print("Stack Overflow! Cannot push")
        else:
            stack.append(int(input("Enter element to push: ")))
            print("Element pushed")

    elif choice == "3":
        if is_empty():
            print("Stack Underflow! Cannot pop")
        else:
            print("Popped element:", stack.pop())

    elif choice == "4":
        if is_empty():
            print("Stack is empty")
        else:
            print("Stack (top to bottom):")
            for i in range(len(stack) - 1, -1, -1):
                print(stack[i])

    elif choice == "5":
        if is_full():
            print("Overflow condition: stack is FULL")
        else:
            print(f"No overflow: {size - len(stack)} space(s) left")

    elif choice == "6":
        if is_empty():
            print("Underflow condition: stack is EMPTY")
        else:
            print(f"No underflow: {len(stack)} element(s) in stack")

    elif choice == "7":
        print("Exiting...")
        break

    else:
```

print("Invalid choice, try again")

Code Screenshot:

```
stack = []
size = 0
created = False

def is_full():
    return len(stack) == size

def is_empty():
    return len(stack) == 0

while True:
    print("\n--- Stack Menu ---")
    print("1. Create a stack")
    print("2. Push an element")
    print("3. Pop an element")
    print("4. Display the stack")
    print("5. Overflow condition")
    print("6. Underflow condition")
    print("7. Exit")

    choice = input("Enter your choice: ")
```

```python
if choice == "1":
    size = int(input("Enter size of the stack: "))
    if size <= 0:
        print("Size must be greater than 0")
    else:
        stack = []
        created = True
        print(f"Stack of size {size} created")

elif choice in ("2", "3", "4", "5", "6") and not created:
    print("Create a stack first (option 1)")

elif choice == "2":
    if is_full():
        print("Stack Overflow! Cannot push")
    else:
        stack.append(int(input("Enter element to push: ")))
        print("Element pushed")

elif choice == "3":
    if is_empty():
        print("Stack Underflow! Cannot pop")
    else:
        print("Popped element:", stack.pop())
```

```
elif choice == "4":
    if is_empty():
        print("Stack is empty")
    else:
        print("Stack (top to bottom):")
        for i in range(len(stack) - 1, -1, -1):
            print(stack[i])

elif choice == "5":
    if is_full():
        print("Overflow condition: stack is FULL")
    else:
        print(f"No overflow: {size - len(stack)} space(s) left")

elif choice == "6":
    if is_empty():
        print("Underflow condition: stack is EMPTY")
    else:
        print(f"No underflow: {len(stack)} element(s) in stack")

elif choice == "7":
    print("Exiting...")
    break

else:
    print("Invalid choice, try again")
```

Code Output:

```
--- Stack Menu ---
1. Create a stack
2. Push an element
3. Pop an element
4. Display the stack
5. Overflow condition
6. Underflow condition
7. Exit
Enter your choice: 1
Enter size of the stack: 5
Stack of size 5 created
```

Queue Question:

Question 1: Write a dynamic program where the user will choose which operation of a queue is to be followed:
1. Creating a new queue
2. Inserting an element into the queue
3. Removing an element from the queue
4. Display
5. Exit

Code:

```
queue = []
size = 0
created = False

while True:
    print("\n--- Queue Menu ---")
    print("1. Create a new queue")
    print("2. Insert an element")
    print("3. Remove an element")
    print("4. Display")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        size = int(input("Enter size of the queue: "))
        if size <= 0:
            print("Size must be greater than 0")
        else:
            queue = []
            created = True
            print(f"Queue of size {size} created")

    elif choice in ("2", "3", "4") and not created:
        print("Create a queue first (option 1)")

    elif choice == "2":
        if len(queue) == size:
            print("Queue Overflow! Cannot insert")
        else:
            queue.append(int(input("Enter element to insert: ")))
```

```
            print("Element inserted")

    elif choice == "3":
        if len(queue) == 0:
            print("Queue Underflow! Cannot remove")
        else:
            print("Removed element:", queue.pop(0))

    elif choice == "4":
        if len(queue) == 0:
            print("Queue is empty")
        else:
            print("Queue (front to rear):", *queue)

    elif choice == "5":
        print("Exiting...")
        break

    else:
        print("Invalid choice, try again")
```

Code Screenshot:

```
queue = []
size = 0
created = False

while True:
    print("\n--- Queue Menu ---")
    print("1. Create a new queue")
    print("2. Insert an element")
    print("3. Remove an element")
    print("4. Display")
    print("5. Exit")

    choice = input("Enter your choice: ")
```

```
if choice == "1":
    size = int(input("Enter size of the queue: "))
    if size <= 0:
        print("Size must be greater than 0")
    else:
        queue = []
        created = True
        print(f"Queue of size {size} created")

elif choice in ("2", "3", "4") and not created:
    print("Create a queue first (option 1)")

elif choice == "2":
    if len(queue) == size:
        print("Queue Overflow! Cannot insert")
    else:
        queue.append(int(input("Enter element to insert: ")))
        print("Element inserted")
```

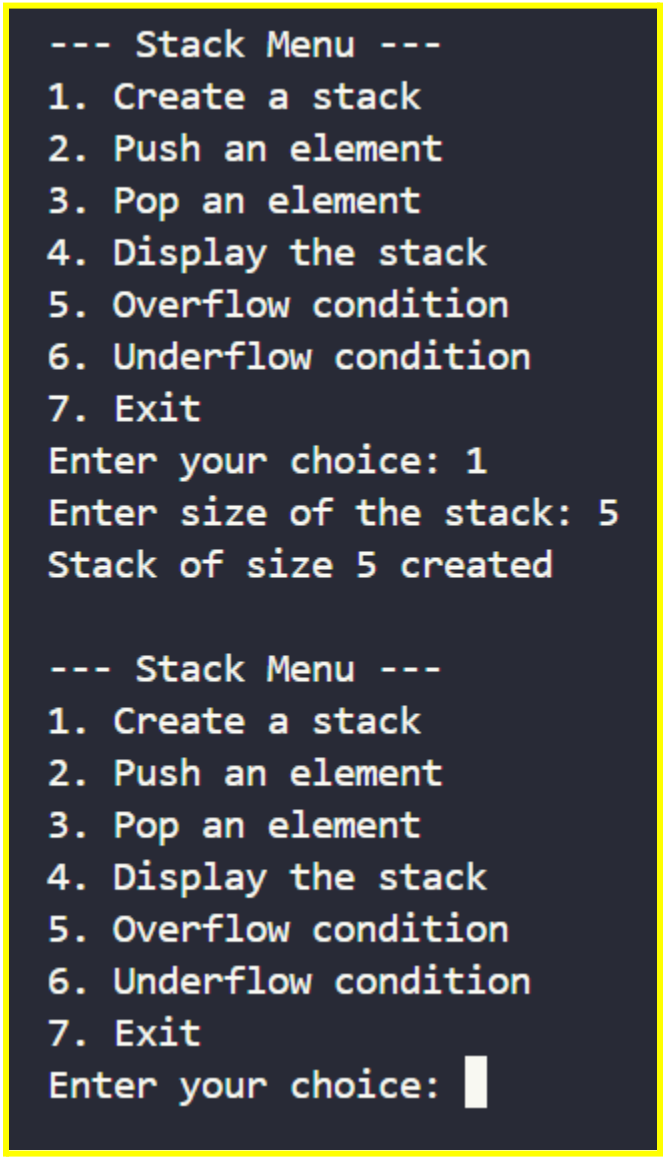
```
elif choice == "3":
    if len(queue) == 0:
        print("Queue Underflow! Cannot remove")
    else:
        print("Removed element:", queue.pop(0))

elif choice == "4":
    if len(queue) == 0:
        print("Queue is empty")
    else:
        print("Queue (front to rear):", *queue)

elif choice == "5":
    print("Exiting...")
    break

else:
    print("Invalid choice, try again")
```

Code Output:


--- Stack Menu ---
1. Create a stack
2. Push an element
3. Pop an element
4. Display the stack
5. Overflow condition
6. Underflow condition
7. Exit
Enter your choice: 1
Enter size of the stack: 5
Stack of size 5 created
--- Stack Menu ---
1. Create a stack
2. Push an element
3. Pop an element
4. Display the stack
5. Overflow condition
6. Underflow condition
7. Exit
Enter your choice: